

Project Design Phase

Problem – Solution Fit Template

Date	15 February 2025
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; Indirectly associated: customers spend free time on volunteering work (I.e. Greenpeace)	Focus on J&P, tap into BE, understand RC
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure > confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	

References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

Purpose of the Canvas

- Solve a real problem experienced by biodiversity researchers, ecologists, and citizen scientists.
 - Increase adoption by building an intuitive, AI-powered butterfly identification tool that aligns with user behaviors.
 - Improve engagement through environmental education and science communication.
 - Build trust with stakeholders by solving a recurring field problem: species misidentification.
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Customer Segment / User Persona

- Field researchers involved in biodiversity tracking
 - Students and educators in environmental sciences
 - Nature photographers and citizen scientists
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The Problem (User Pain Points)

- Manual butterfly identification is time-consuming and error-prone.
 - Non-experts (e.g., students, public) lack tools for accurate species recognition.
 - Existing apps are either outdated, inaccurate, or not inclusive of diverse species (like 75 classes).
 - Researchers lose time cross-referencing field guides or expert opinions.
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The Solution

- A transfer learning-based image classification system that accurately identifies butterfly species from images.
- Fast, mobile-friendly predictions using a simple web interface (Streamlit-based).
- Trained on 6499+ images across 75 butterfly classes to ensure generalization.

- Real-time accuracy, visualization, and classification feedback for both experts and beginners.

 Key Features

- VGG16 or ResNet-based transfer learning architecture
- Model accuracy ~93% with fine-tuning
- Streamlit or Flask frontend for user interaction
- Educational output with species name, confidence score, and ecological relevance

 Problem–Solution Fit

Component	Details
User Behavior	Captures butterflies, wants instant ID, contributes to science
Problem Encountered	No accurate, fast, or field-friendly classification tool available
Solution Offered	AI model that works offline/online and simplifies classification
Behavior Match	Users already use cameras, mobile tools – the solution extends their habit
Value Proposition	Saves time, increases ID accuracy, supports conservation with meaningful data